

A Note on Pairing for Beginners

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Abstract

This note complements the work [Cos12], which provides a detailed introduction to the Weil pairing and Tate pairing, by more details and proofs omitted in the original work.

1 Preliminaries and Assumptions

In this work, we assume the following:

- $E = E(\overline{\mathbb{F}_q})$ is an elliptic curve defined over $\overline{\mathbb{F}_q}$, where q is a prime number > 3 . $E(K)$ specifies the set of points on E defined over a subfield K of $\overline{\mathbb{F}_q}$.
- r is a prime number such that $r \nmid q$ and $r \mid \#E(\mathbb{F}_q)$. We also assume $r^2 \nmid \#E(\mathbb{F}_q)$. We write $r \parallel \#E(\mathbb{F}_q)$ if $r \mid \#E(\mathbb{F}_q)$ but $r^2 \nmid \#E(\mathbb{F}_q)$. $E[r]$ is the r -torsion subgroup of E , which is defined as $E[r] = \{P \in E : [r]P = \mathcal{O}\}$. Since r is coprime to q , we have $E[r] \cong \mathbb{Z}_r \times \mathbb{Z}_r$.
- k is the embedding degree of E with respect to r , i.e., k is the smallest positive integer such that $r \mid q^k - 1$. Since $r \parallel \#E(\mathbb{F}_q)$, we have the embedding degree $k > 1$.
- For a function f on E , we denote the divisor of f as (f) .
- $\text{Div}_{\overline{\mathbb{F}_q}}(E)$ is the group of divisors on $E(\overline{\mathbb{F}_q})$. $\text{Div}^0(E)$ is the group of degree-zero divisors in $\text{Div}(E)$. $\text{Prin}(E)$ is the group of principal divisors in $\text{Div}(E)$. Note that $\text{Prin}(E) \subset \text{Div}^0(E) \subset \text{Div}(E)$. We define the divisor class group, or Picard group of E by

$$\text{Pic}^0(E) = \text{Div}^0(E)/\text{Prin}(E)$$

- Two divisors D_1 and D_2 are called equivalent, written as $D_1 \sim D_2$, if $D_1 = D_2 + (f)$ for some function f .
- Given $m \in \mathbb{Z}$ and $P \in E$, define the function $f_{m,P}$ as the function whose divisor is $(f_{m,P}) = m(P) - ([m]P) - (m-1)(\mathcal{O})$.
 - For $m = 0, 1$, $(f_{m,P})$ is the zero divisor.
 - For $P \in E[r]$, $(f_{r,P}) = r(P) - r(\mathcal{O})$.

Note that we have

$$(f_{m+1,P}) - (f_{m,P}) = (P) + ([m]P) - ([m+1]P) - (\mathcal{O}) = \left(\frac{\ell_{[m]P,P}}{v_{[m+1]P}} \right)$$

where

- $\ell_{[m]P,P}$: the sloped line connecting P and $[m]P$. $(\ell_{[m]P,P}) = (P) + ([m]P) + ([m+1]P) - 3(\mathcal{O})$.
- $v_{[m+1]P}$: the vertical line passing through $[m+1]P$. $(v_{[m+1]P}) = 2([m+1]P) - 2(\mathcal{O})$.
- μ_r is the group of r -th roots of unity in $\overline{\mathbb{F}_q}$, i.e., $\mu_r = \{x \in \overline{\mathbb{F}_q}^* : x^r = 1\}$.

We recall a few important definitions and theorems.

Theorem 1 (Weil Reciprocity). *Let f and g be non-zero functions on a curve such that (f) and (g) have disjoint supports. Then $f((g)) = g((f))$.*

Theorem 2. If E is defined over a field with characteristic 0 or prime to r , then

$$E[r] \cong \mathbb{Z}_r \times \mathbb{Z}_r$$

Corollary 1. If E is defined over a field with characteristic 0 or prime to r , then

- $\#E[r] = r^2$.
- $E[r]$ is a $\mathbb{F}_r$ -vector space of dimension 2.
- If r is a prime number, $E[r]$ has $r + 1$ cyclic subgroups of order r

Proof. The first two properties are straightforward from Theorem 2. For the last property, since r is prime, all subgroups of order r don't share any non-zero element. Indeed, if two subgroups share any non-zero elements g , then both subgroups are $\langle g \rangle$. In addition, for any subgroup of order r , any element is a generator. $r^2 - 1 = (r - 1)(r + 1)$, so there are $r + 1$ different subgroups of order $r - 1$. $\square$

Theorem 3. Let r be a prime number, $r \neq q$ and $r \mid \#E(\mathbb{F}_q)$. The embedding degree k is the smallest integer such that $E[r] \subset E(\mathbb{F}_{q^k})$.

Proof. Recall the Frobenius map $\pi(x, y) = (x^q, y^q)$, which holds that $P \in E(\mathbb{F}_{q^k}) \Leftrightarrow \pi^k(P) = P$ for any $k \in \mathbb{Z}$. We show that $r \mid q^k - 1$ if and only if any point $P \in E[r]$ satisfies $\pi^k(P) = P$.

Since $E[r] \cong \mathbb{Z}_r \times \mathbb{Z}_r$ and π is a linear transformation ($\pi(P) + \pi(Q) = \pi(P + Q)$) we can write $E[r]$ as a vector space and π as a matrix. The characteristic function is

$$X^2 - tX + q = 0$$

Since $r \mid \#E(\mathbb{F}_q) = q + 1 - t$, $t \bmod r = q + 1$. The characteristic function modulo r becomes

$$X^2 - (q + 1)X + q = 0 \bmod r \iff (X - 1)(X - q) = 0 \bmod r.$$

So there are two eigenvalues: 1 and q .

Suppose $k > 1$, $r \nmid q - 1$, the eigenvalues are distinct, and $E[r]$ is split into the sum of two 1-dimensional space.

- Eigenvalue 1: These are points P such that $\pi(P) = P$, which implies $\pi^k(P) = P$.
- Eigenvalue q : These are points P such that $\pi(P) = [q]P$. Hence, $\pi^k(P) = [q^k]P$.

Since P is in the r -torsion, we also have $[r]P = \mathcal{O}$. If $r \mid q^k - 1$, then $\pi^k(P) = P$; on the other hand, if $\pi^k(P) = P$, then $q^k = 1 \bmod r$.

Suppose $k = 1$, then all points $P \in E[r]$ satisfies $\pi(P) = P$, and thus $E[r] \subset E(\mathbb{F}_q)$. $\square$

Definition 1 (Trace Map). The trace map $\text{Tr} : E(\mathbb{F}_{q^k}) \rightarrow E(\mathbb{F}_q)$ is defined as

$$\text{Tr}(P) = \sum_{\sigma \in \text{Gal}(\mathbb{F}_{q^k}/\mathbb{F}_q)} \sigma(P) = \sum_{i=0}^{k-1} \pi^i(P) = \sum_{i=0}^{k-1} (x^{q^i}, y^{q^i})$$

We can also define the anti-trace map by $\text{aTr}(P) = [k]P - \text{Tr}(P)$. Note that for all $P \in E(\mathbb{F}_q)$, $\text{aTr}(P) = \mathcal{O}$; for all $P \in E(\mathbb{F}_{q^k})$, $\text{Tr}(\text{aTr}(P)) = \mathcal{O}$.

Proposition 1.

- Tr is a group homomorphism.
- Tr sends all r -torsion points into one subgroup of r -torsion.

A nice illustration is given in Figure 1.

Now we emphasize the findings from the proof of Theorem 3, which characterizes the structure of $E[r]$. There exists a unique subgroup of $E[r]$ in $E(\mathbb{F}_q)$, which is the $[1]$ -eigenspace of π , or $E[r] \cap \ker(\pi - [1])$. We call this subgroup the *base-field subgroup* $\mathcal{G}_1$.

- For any $P \in \mathcal{G}_1$, $\text{Tr}(P) = [k]P$. One can check it by $\sum_{i=0}^{k-1} 1 = k$.

1.1 Pairing Types

Type-1 Pairing.

- E is supersingular equipped with the distortion map ϕ
- $\mathbb{G}_1 = \mathbb{G}_2 = \mathcal{G}_1$
- $e(P, Q) = \hat{e}(P, \phi(Q))$, where $\hat{e}$ is the Weil or Tate pairing.
- No hashing problem: getting into $E(\mathbb{F}_q)[r]$ requires a simple cofactor multiplication once we have hashed into $E(\mathbb{F}_q)$.
- Type-1 pairing is not very efficient: supersingular curves are highly restrictive when it comes to optimization.

Type-2 Pairing.

- $\mathbb{G}_1 = \mathcal{G}_1$ and $\mathbb{G}_2$ is any of the $r - 1$ subgroups in $E[r]$ that is not $\mathcal{G}_1$ or $\mathcal{G}_2$.
- There exists a map $\psi : \mathbb{G}_2 \rightarrow \mathbb{G}_1$ as the trace map.
- There is no known way of hashing into $\mathbb{G}_2$ or generating random elements of $\mathbb{G}_2$.
- We can still specify a generator $P_2 \in \mathbb{G}_2$ and generate elements via scalar multiplications. This is often undesirable though since we cannot generate random elements without knowing the discrete logarithm.

Type-3 Pairing.

- $\mathbb{G}_1 = \mathcal{G}_1$ and $\mathbb{G}_2 = \mathcal{G}_\infty$.
- We can hash into $\mathbb{G}_2$. One way is by hashing into $E(\mathbb{F}_{q^k})$, a cofactor multiplication into $E(\mathbb{F}_{q^k})[r]$, and the map $\text{aTr} : E[r] \rightarrow \mathbb{G}_2$. But we have a more efficient way.
- An isomorphism $\psi : \mathbb{G}_2 \rightarrow \mathbb{G}_1$ exists, but we do not have an efficient way to compute it.

Type-4 Pairing.

- $\mathbb{G}_1 = \mathcal{G}_1$ and $\mathbb{G}_2 = E[r]$.
- Hashing into $\mathbb{G}_2$ is doable but not very efficient.
- We cannot hash into $\mathbb{G}_2$ generated by some P_2 since $\mathbb{G}_2$ is not cyclic now.

Unless otherwise stated, we consider Type-3 pairings: $\mathbb{G}_1 = \mathcal{G}_1 = E[r] \cap \ker(\pi - [1])$ and $\mathbb{G}_2 = \mathcal{G}_2 = E[r] \cap \ker(\pi - [q])$, which is usually the most efficient choice.

1.2 Twisted Curves

A twist of $E : y^2 = x^3 + ax + b$ is given by $E' : y^2 = x^3 + a\omega^4x + b\omega^6$, with the map

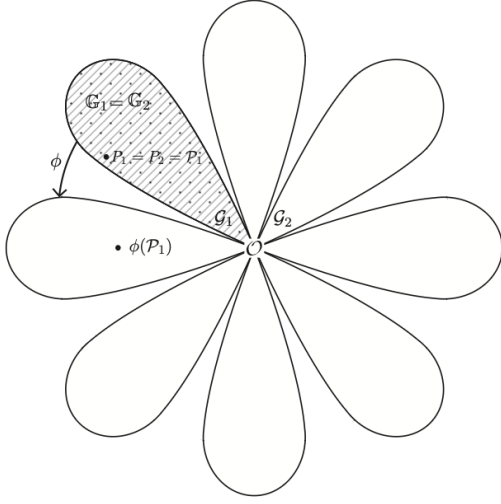
$$\Psi : E'(\mathbb{F}_{q^{k/d}}) \rightarrow E(\mathbb{F}_{q^k}) : (x', y') \mapsto (x'/\omega^2, y'/\omega^3)$$

where $\omega \in \mathbb{F}_{q^k}$. We represent elements Q in $\mathbb{G}_2 = \mathcal{G}_2 \in E(\mathbb{F}_{q^k})$ as points Q' in a smaller field $\mathbb{F}_{q^{k/d}}$, where the embedding degree k has d as a factor.

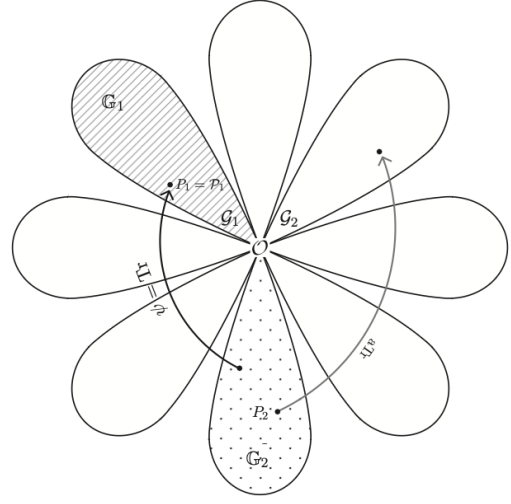
The only available degrees of the twist on elliptic curves are $d \in \{2, 3, 4, 6\}$.

$d = 2$ Quadratic Twists.

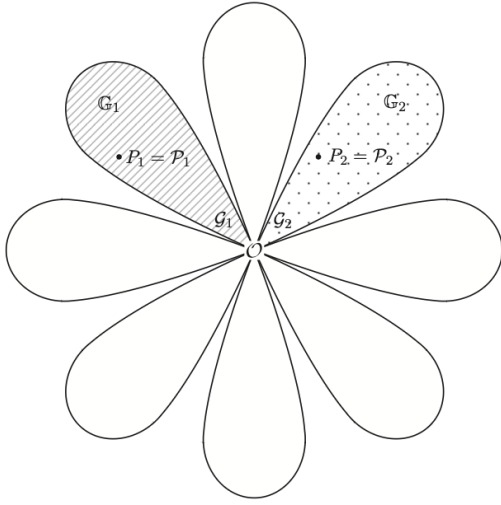
- Quadratic twists are available on any elliptic curve.
- If $E/\mathbb{F}_q : y^2 = x^3 + ax + b$, then $E'/\mathbb{F}_{q^{k/2}} : y^2 = x^3 + a\omega^4x + b\omega^6$, with $\omega^2 \in \mathbb{F}_{q^{k/2}}$.
- Ψ takes elements in $E'(\mathbb{F}_{q^{k/2}})$ to elements in $E(\mathbb{F}_{q^k})$.



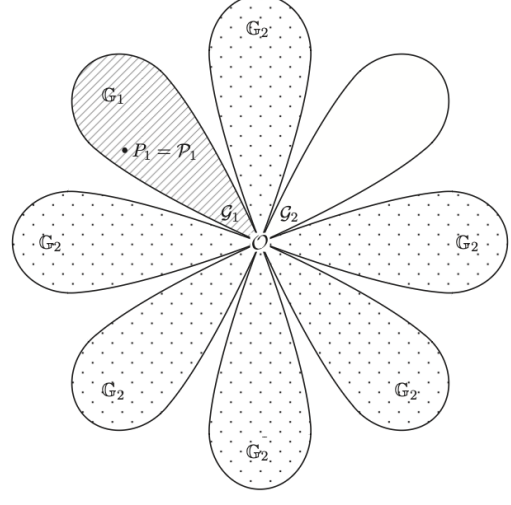
(a) Type 1 pairings.



(b) Type 2 pairings.



(c) Type 3 pairings.



(d) Type 4 pairings.

Figure 2: The four types of pairings (from [Cos12]).

$d = 3$ Cubic Twists.

- Cubic twists are only available when $a = 0$.
- If $E/\mathbb{F}_1 : y^2 = x^3 + b$, then $E'/\mathbb{F}_{q^{k/2}} : y^2 = x^3 + b\omega^6$, with $\omega^3, \omega^6 \in \mathbb{F}_{q^{k/3}}$ but $\omega^2 \in \mathbb{F}_{q^k} \setminus \mathbb{F}_{q^{k/3}}$.
- Ψ takes elements in $E'(\mathbb{F}_{q^{k/3}})$ to elements in $E(\mathbb{F}_{q^k})$.

$d = 4$ Quartic Twists.

- Quartic twists are only available when $b = 0$.
- If $E/\mathbb{F}_1 : y^2 = x^3 + ax$, then $E'/\mathbb{F}_{q^{k/4}} : y^2 = x^3 + a\omega^4x$ with $\omega^4 \in \mathbb{F}_{q^{k/4}}$ and $\omega^2 \in \mathbb{F}_{q^{k/2}}$ but $\omega^3 \in \mathbb{F}_{q^k} \setminus \mathbb{F}_{q^{k/2}}$.
- Ψ takes elements in $E'(\mathbb{F}_{q^{k/4}})$ to elements in $E(\mathbb{F}_{q^k})$.

$d = 6$ Sextic Twists.

- Sextic twists are only available when $a = 0$.
- If $E/\mathbb{F}_1 : y^2 = x^3 + b$, then $E'/\mathbb{F}_{q^{k/6}} : y^2 = x^3 + b\omega^6$, with $\omega^6 \in \mathbb{F}_{q^{k/6}}$ and $\omega^3 \in \mathbb{F}_{q^{k/3}}$ and $\omega^2 \in \mathbb{F}_{q^{k/2}}$.
- Ψ takes elements in $E'(\mathbb{F}_{q^{k/6}})$ to elements in $E(\mathbb{F}_{q^k})$.

2 Weil Pairing

Definition 3 (Weil Pairing). Let $P, Q \in E(\mathbb{F}_{q^k})[r]$ and let $D_P, D_Q \in \text{Div}^0(E)$ be degree-zero divisors such that $D_P \sim (P) - (\mathcal{O})$, $D_Q \sim (Q) - (\mathcal{O})$, and D_P and D_Q have disjoint supports. There exist functions f and g such that $(f) = rD_P$ and $(g) = rD_Q$. The Weil pairing $w_r : E(\mathbb{F}_{q^k})[r] \times E(\mathbb{F}_{q^k})[r] \rightarrow \mu_r$ is defined as

$$w_r(P, Q) = \frac{f(D_Q)}{g(D_P)}.$$

Well-definedness.

- Free choice of f : Given $D \in \text{Div}^0$, the function f such that $(f) = D$ is unique up to a scalar multiple $\lambda \in \mathbb{F}_{q^k}^*$. For any λf , $(\lambda f)(D_Q) = \lambda^{\deg D_Q} \cdot f(D_Q) = f(D_Q)$ since $\deg D_Q = 0$.
- Free choice of divisor: If we replace some D_P with an equivalent D'_P , then $D'_P = D_P + (h)$ for some function h , and we would have f' be such that $(f') = rD'_P$, so $f' = f \cdot h^r$. Now, we have

$$f'(D_Q) = f(D_Q) \cdot h(D_Q)^r, \quad g(D'_P) = g(D_P) \cdot g((h))$$

By Weil Reciprocity, $g((h)) = h((g)) = h(rD_Q) = h(D_Q)^r$. Hence $\frac{f(D_Q)}{g(D_P)} = \frac{f'(D_Q)}{g(D'_P)}$.

- Range is the unity: $\omega_r(P, Q)^r = \frac{f^r(D_Q)}{g^r(D_P)} = \frac{f(rD_Q)}{g(rD_P)}$. Note that $(f) = rD_P$ and $(g) = rD_Q$, so $w_r(P, Q)^r = \frac{f((g))}{g((f))} = 1$ by Weil Reciprocity.

2.1 Evaluation of Weil Pairing

Pick $D'_P = (P) - (\mathcal{O})$, $f = f_{r,P}$, $D_Q = (Q + S) - (S)$ for some random point S , and $g = f_{r,Q} \circ T_{-S}$, where $T_{-S}(X) = X - S$ shifts a point X to $X - S$. Note that $(f) = r(P) - r(\mathcal{O}) = rD'_P$. Moreover, if $f_{r,Q}$ has a zero or pole at a point X , g has a zero or pole at the point $X + S$. As $(f_{r,Q}) = r(Q) - r(\mathcal{O})$, we also have $(g) = r(Q + S) - r(S)$.

The Weil pairing evaluates to

$$w_r(P, Q) = \frac{f(D_Q)}{g(D'_P)} = \frac{f_{r,P}(D_Q)}{f_{r,Q}(P - S)/f_{r,Q}(-S)} = \frac{f_{r,P}(D_Q)}{f_{r,Q}(D_P)},$$

where $D_P = (P - S) - (-S)$. Therefore, the Weil pairing can be evaluated by Miller's algorithm, which computes $f_{r,P}(D_Q)$ and $f_{r,Q}(D_P)$ in $O(\log r)$ steps.

3 Tate Pairing

Recall that we assume $r \nmid \#E(\mathbb{F}_q)$. We now also assume that $r^2 \nmid \#E(\mathbb{F}_{q^k})$.

Define

$$rE(\mathbb{F}_{q^k}) = \{[r]P : P \in E(\mathbb{F}_{q^k})\}$$

This is a subgroup of $E(\mathbb{F}_{q^k})$ and has order $\frac{\#E(\mathbb{F}_{q^k})}{r^2} = h$. Therefore, we can consider the quotient group

$$E/rE : \{R + rE : R \in E(\mathbb{F}_{q^k})\}, \quad P_1 + rE = P_2 + rE \Leftrightarrow P_1 - P_2 \in rE$$

There are r^2 distinct cosets, each has a unique element in $E[r]$.

More Details. Consider the map $[r] : P \mapsto [r]P$. It is a homomorphism and has kernel $E[r]$ and image rE . By the first isomorphism theorem,

$$E(\mathbb{F}_{q^k})/E[r] \cong rE(\mathbb{F}_{q^k}).$$

Hence, $\#rE = \frac{\#E(\mathbb{F}_{q^k})}{r^2}$. Moreover, since $r^3 \nmid \#E(\mathbb{F}_{q^k})$, we can split $E(\mathbb{F}_{q^k}) = E[r] \oplus E_c$ for some subgroup E_c of order c by the Fundamental Theorem of Finite Abelian Groups. If $R \in rE$, then $R = (rA, rB)$ for some $(A, B) \in (E[r], E_c)$, which means $R = (0, rB)$. Since c is coprime to r , the set $\{rB : B \in E_c\} = E_c$. rE is exactly E_c . Therefore, $rE \cap E[r] = \{\mathcal{O}\}$.

This means we also have

$$E(\mathbb{F}_{q^k})/rE(\mathbb{F}_{q^k}) \cong E[r].$$

Therefore, each $P \in E[r]$ yields a distinct $P + rE$. There are r^2 elements in $E[r]$, and there are also r^2 cosets.

Definition 4 (Tate Pairing). Let $P \in E(\mathbb{F}_{q^k})[r]$ and let f be a function whose divisor is $(f) = r(P) - r(\mathcal{O})$. Let $Q \in E(\mathbb{F}_{q^k})/rE(\mathbb{F}_{q^k})$ and let D_Q be a degree-zero divisor that is equivalent to $(Q) - (\mathcal{O})$ and has a support disjoint to that of (f) . The Tate pairing $t_r : E(\mathbb{F}_{q^k})[r] \times E(\mathbb{F}_{q^k})/rE(\mathbb{F}_{q^k}) \rightarrow \mathbb{F}_{q^k}^*/(\mathbb{F}_{q^k}^*)^r$ is defined as

$$t_r(P, Q) = f(D_Q).$$

We note that $(\mathbb{F}_{q^k}^*)^r = \{u^r : u \in \mathbb{F}_{q^k}^*\}$.

Well-definedness.

- Free choice of f : Given a divisor D . If $(f) = D$, then f is unique up to a non-zero scalar multiple. Since $\deg D_Q = 0$, $f(D_Q)$ is fixed.
- Free choice of divisor: Let D_Q, D'_Q be two divisors equivalent to $(Q) - (\mathcal{O})$ and have disjoint support to that of (f) . There exists a function h such that $D'_Q = D_Q + (h)$. Hence, $f(D'_Q) = f(D_Q) \cdot f((h))$. By Weil Reciprocity, $f((h)) = h((f)) = (\frac{h(P)}{h(\mathcal{O})})^r$. $f(D'_Q)$ is equivalent to $f(D_Q)$ up to a multiple in $(\mathbb{F}_{q^k}^*)^r$.
- Image in $\mathbb{F}_{q^k}^*/(\mathbb{F}_{q^k}^*)^r$: Let $Q' = Q + [r]S$ for some $S \in E(\mathbb{F}_{q^k})$. We show that $t_r(P, Q) = t_r(P, Q') \bmod (\mathbb{F}_{q^k}^*)^r$. Let D'_Q be a degree-zero divisor that is equivalent to $(Q') - (\mathcal{O})$ and has a disjoint support to (f) . Let D_S be a divisor equivalent to $(S) - (\mathcal{O})$ and has a disjoint support to (f) .

$$(Q') - (Q) \sim (Q') - (Q) - [(Q') - (Q) - r(S) + r(\mathcal{O})] \sim rD_S$$

Therefore,

$$(Q') - (\mathcal{O}) \sim (Q) - (\mathcal{O}) + rD_S \implies D'_Q \sim D_Q + rD_S$$

Let $D'_Q = D_Q + rD_S + (h)$ for some function h . Again, by Weil Reciprocity

$$f(D'_Q) = f(D_Q) \cdot f(D_S)^r \cdot f((h)) = f(D_Q) \cdot f(D_S)^r \cdot \left(\frac{h(P)}{h(\mathcal{O})}\right)^r$$

Therefore, $t_r(P, Q) = t_r(P, Q') \bmod (\mathbb{F}_{q^k}^*)^r$.

Since the output lies in an equivalence class $\mathbb{F}_{q^k}^*/(\mathbb{F}_{q^k}^*)^r$, it is not useful in practice unless we modify the definition.

Definition 5 (Reduced Tate Pairing). Let $P \in E(\mathbb{F}_{q^k})[r]$ and let f be a function whose divisor is $(f) = r(P) - r(\mathcal{O})$. Let $Q \in E(\mathbb{F}_{q^k})/rE(\mathbb{F}_{q^k})$ and let D_Q be a degree-zero divisor that is equivalent to $(Q) - (\mathcal{O})$ and has a support disjoint to that of (f) . The *reduced* Tate pairing $t_r : E(\mathbb{F}_{q^k})[r] \times E(\mathbb{F}_{q^k})/rE(\mathbb{F}_{q^k}) \rightarrow \mu_r$ is defined as

$$T_r(P, Q) = t_r(P, Q)^{(q^k-1)/r} = f(D_Q)^{(q^k-1)/r}.$$

One can just use $f = f_{r,P}$.

Well-definedness of Reduced Tate Pairing.

- Same elements in $\mathbb{F}_{q^k}^*/(\mathbb{F}_{q^k}^*)^r$ mapped to the same elements in μ_r : Given $u, v \in \mathbb{F}_{q^k}^*$ such that $u = \lambda v, \lambda \in (\mathbb{F}_{q^k}^*)^r$. We have $u^{(q^k-1)/r} = v^{(q^k-1)/r} \cdot \lambda^{(q^k-1)/r} = v^{(q^k-1)/r}$.
- Same elements in μ_r correspond to the same elements in $\mathbb{F}_{q^k}^*/(\mathbb{F}_{q^k}^*)^r$: Let $u = u_r^{(q^k-1)/r}$ and $v = v_r^{(q^k-1)/r}$ and $u = v$. Suppose $u_r = \lambda v_r$, then $\lambda^{(q^k-1)/r} = 1$. Since $\mathbb{F}_{q^k}^*$ is a cyclic group, there exists a generator g such that $g^\ell = \lambda$ for some ℓ . So we have $q^k - 1 \mid \frac{q^k-1}{r} \cdot \ell$, which implies ℓ is a multiple of r and $\lambda = (g^{\ell/r})^r \in (\mathbb{F}_{q^k}^*)^r$. Therefore, $u_r = v_r \bmod (\mathbb{F}_{q^k}^*)^r$.

We have shown that each element in $E[r]$ corresponds to one element in $E(\mathbb{F}_{q^k})/rE(\mathbb{F}_{q^k})$. Therefore, from now on we will always consider $P \in \mathcal{G}_1 = E[r] \cap \ker(\pi - [1])$ and $Q \in \mathcal{G}_2 = E[r] \cap \ker(\pi - [q])$.

4 Miller's Algorithm

We recall that the function $f_{m,P}$ is defined by its divisor $(f_{m,P}) = m(P) - ([m]P) - (m-1)(\mathcal{O})$. We have seen that $f_{m+1,P} = f_{m,P} \cdot \frac{\ell_{[m]P,P}}{v_{[m+1]P}}$ and $f_{1,P} = 1$.

For Weil or Tate pairing, we need to compute $f_{r,P}$, where $P \in E[r]$. Note that (v_{rP}) is the zero divisor, $\ell_{[r-1]P,P}$ is the vertical line in the denominator of the previous term $\frac{\ell_{[r-2]P,P}}{v_{[r-1]P}}$. Hence

$$f_{r,P} = \ell_{[r-2]P,P} \cdot \prod_{i=1}^{r-3} \frac{\ell_{iP,P}}{v_{[i+1]P}}$$

Therefore, the computation requires a complexity $O(r)$. For exponential r , it is very inefficient.

Instead of adding one zero at P and one pole at $\mathcal{O}$, we can *double*. That is, observe that

$$\begin{aligned} (f_{2m,P}) &= 2m(P) - ([2m]P) - (2m-1)(\mathcal{O}), (f_{m,P}^2) = 2m(P) - 2([m]P) - (2m-2)(\mathcal{O}) \\ \implies (f_{2m,P}) - (f_{m,P}^2) &= 2([m]P) - ([2m]P) - (\mathcal{O}) = \left(\frac{\ell_{[m]P,[m]P}}{v_{[2m]P}} \right) \end{aligned}$$

where $\ell_{[m]P,[m]P}$ is the tangent line at $[m]P$. We can thus update $f_{2m,P} = f_{m,P}^2 \cdot \frac{\ell_{[m]P,[m]P}}{v_{[2m]P}}$.

Finally, since $f_{r,P}$ has a degree as large as r , so we do not compute $f_{r,P}$, but evaluate $f_{m,P}(D_Q) \in \mathbb{F}_{q^k}$ at each stage.

Algorithm 1 Miller's Algorithm

Input: $P \in \mathbb{E}(\mathbb{F}_{q^k})[r]$, D_Q with support disjoint from $f_{r,P}$, $r = (r_{n-1} \dots r_1 r_0)_2$ with $r_{n-1} = 1$.

Output: $f_{r,P}(D_Q)$

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1:  $R \leftarrow P, f \leftarrow 1$ 
2: for  $i = n-2$  down to 0 do
3:   Compute  $\ell_{R,R}$  and  $v_{[2]R}$ 
4:    $f \leftarrow f^2 \cdot \frac{\ell_{R,R}}{v_{[2]R}}(D_Q)$ 
5:    $R \leftarrow [2]R$ 
6:   if  $r_i = 1$  then
7:     Compute  $\ell_{R,P}$  and  $v_{R+P}$ 
8:      $f \leftarrow f \cdot \frac{\ell_{R,P}}{v_{R+P}}(D_Q)$ 
9:      $R \leftarrow R + P$ 
10:  end if
11: end for
12: return  $f$ 
```

4.1 Optimizations

For Tate pairing, we can enjoy many optimizations.

- If P and Q are independent, $f_{r,P}(D_Q)^{(q^k-1)/r} = f_{r,P}(Q)^{(q^k-1)/r}$. This not only eliminates the need to find an equivalent divisor D_Q that has a different support than $(Q) - (\mathcal{O})$ but also to evaluate f at two points.
- We have $q-1 \mid \frac{q^k-1}{r}$. This is because $k > 1$ and thus $r \nmid q-1$. Let $q^k-1 = c \cdot (q-1)$. Since $f_{r,P}(Q)^{q^k-1} = (f_{r,P}(Q)^{q-1})^c$, we can multiply or divide $f_{r,P}(Q)$ by any factors in $\mathbb{F}_q$ without affecting the final pairing value.
 - If E is supersingular, $k = 2$, and the x -coordinate of Q is always in $\mathbb{F}_q$. As a result, we can remove all the $v_{[2]R}(Q)$ computation in the Miller's algorithm.
 - If E is ordinary, suppose k is even, x_Q is always in $\mathbb{F}_{q^{k/2}}$ by Lemma 1. We can also remove the denominator in this case as $q^{k/2}-1 \mid \frac{q^k-1}{r}$.

Lemma 1. Let $Q \in \mathcal{G}_2 = E[r] \cap \ker(\pi - [q])$. If k is even, then the x -coordinate of Q is in $\mathbb{F}_{q^{k/2}}$.

Proof. Suppose k is even, $q^k - 1 = (q^{k/2} - 1)(q^{k/2} + 1)$. Since $r \nmid q^{k/2-1}$ and $r \mid q^k - 1$, we have $q^{k/2} + 1 = 0 \pmod r$. Notice that $\pi^{k/2}(Q) = (x_Q^{q^{k/2}}, y_Q^{q^{k/2}}) = [q^{k/2}]Q = -Q = (x_Q, -y_Q)$. The first equality is by definition of π , and the second is because Q is in $[q]$ -eigenspace of π . Now, $x_Q^{q^{k/2}} = x_Q$ implies $x_Q \in \mathbb{F}_{q^{k/2}}$. □

Algorithm 2 Miller's Algorithm (The BKLS-GHS Version)

Input: $P \in \mathbb{G}_1$, $Q \in \mathbb{G}_2$ on some twist of degree $d \mid k$, k is even, $r = (r_{n-1} \dots r_1 r_0)_2$ with $r_{n-1} = 1$

Output: $f_{r,P}(Q)^{(q^k-1)/r}$

```

1:  $R \leftarrow P, f \leftarrow 1$ 
2: for  $i = n - 2$  down to  $0$  do
3:   Compute  $\ell_{R,R}$ 
4:    $f \leftarrow f^2 \cdot \ell_{R,R}(Q)$ 
5:    $R \leftarrow [2]R$ 
6:   if  $r_i = 1$  then
7:     Compute  $\ell_{R,P}$ 
8:      $f \leftarrow f \cdot \ell_{R,P}(Q)$ 
9:      $R \leftarrow R + P$ 
10:  end if
11: end for
12: return  $f^{(q^k-1)/r}$ 

```

We note that in practice, we can use projective coordinates to represent points on the elliptic curve to further optimize the computation by avoiding expensive field inversions.

- Since Q is fixed during the Miller loop, we do not need projective coordinates for it.
- Compute $R \leftarrow [2]R$ and $R \leftarrow R + P$ in projective coordinates.
- We can also compute $\ell_{R,R}(\cdot)$ and $\ell_{R,P}(\cdot)$ in projective coordinates.

4.2 Role of Twists in Miller's Algorithm

Mathematically, a point $Q \in \mathbb{G}_2$ on the original curve $E(\mathbb{F}_{q^k})$ and its twisted counterpart $Q' = \Psi^{-1}(Q)$ on the smaller curve $E'(\mathbb{F}_{q^{k/d}})$ represent the exact same geometric data. However, in a computer's memory, finite field elements are represented as polynomials. Using a twist functions as an optimized *change of variables* (or change of basis) to an algebraic structure that is mostly zeros.

- Without a twist:
 1. Evaluating $\ell(Q)$ in Miller's algorithm yields a *dense* polynomial in $\mathbb{F}_{q^k}$.
 2. Running accumulator: $f \leftarrow f^2 \cdot \ell(Q)$ requires a *dense-by-dense* field multiplication (e.g., $k \times k$ base field cross-multiplications).
- With a twist:
 1. Evaluating $\ell(\Psi(Q'))$ yields a *sparse* polynomial where only a few coefficients (associated with specific powers of the twist variable ω) are non-zero.
 2. Running accumulator: $f \leftarrow f^2 \cdot \ell(Q)$ requires a *dense-by-sparse* field multiplication (e.g., $k \times k/d$ base field cross-multiplications).

References

- [Cos12] Craig Costello. *Pairings for Beginners*. <https://static1.squarespace.com/static/5fdbb09f31d71c1227082339/t/5ff394720493bd28278889c6/1609798774687/PairingsForBeginners.pdf>. 2012.